

Cargo Space

Comprehension questions 1

The unfilled space in the tanks or other receptacles carrying liquids is called **<ullage>**.

The empty space which is not taken up by the cargo is usually called **<broken stowage>**.

Grain capacity is greater than bale capacity because **<grain fills the holds entirely>** and there is no space left unfilled.

Cargo space on board ship can be measured in cubic metres or **<cubic feet>**.

General cargo is usually **<packed>** in different kind of packages and bulk cargo is loose and unpacked.

Cargo Space

Comprehension questions 2

It is easier to stow containers because they are of standard shape and **<dimensions>**.

Sulphur can be carried in a liquid or molten form by specially constructed **<chemical carriers>**.

Perishable cargo includes fish, meat, eggs, **<dairy produce>** such as butter, cheese and milk and different kinds of fruit.

Gases can be transported by **<LNG and LPG vessels>**.

Ore can be carried in the hold of **<OBO ships>**.

Cargo Space

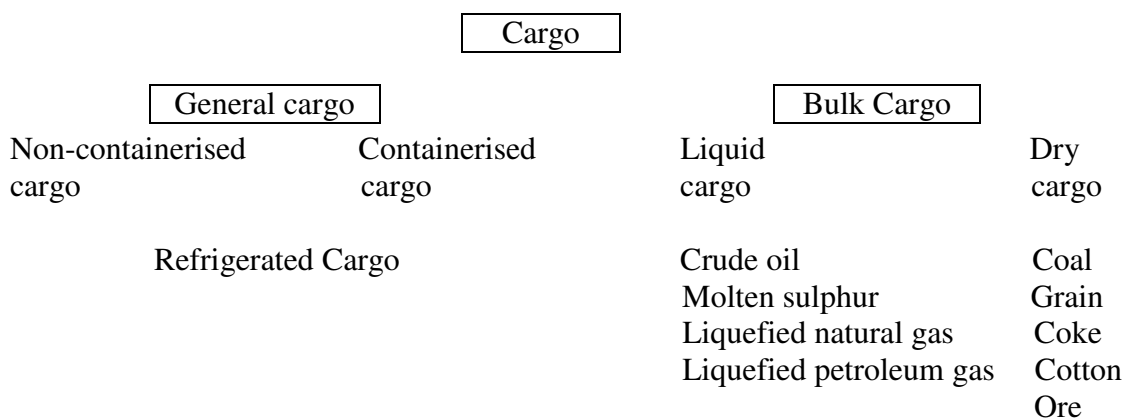
Exercise 1

1. FALSE
2. TRUE
3. FALSE
4. TRUE
5. TRUE

Exercise 2

	-t	-d	-id
expressed	X		
measured		X	
packed	X		
divided			X
transported			X
filled		X	
loaded			X
locked	X		
inspected			X
discharged		X	

Exercise 3



Exercise 4

ROPAX	Passenger and car ferry
CON-RO	Container and Ro-Ro vessel
Broken Stowage	Unfilled space between packages and pieces of machinery
TEU	Twenty-Foot Equivalent Unit
OBO	Ore/Bulk/Oil
Reefer	Refrigerated vessel
Lo-Lo	Lift on-Lift off
Ullage	Unfilled space of a tank or other receptacle containing liquid
PCC	Pure Car Carrier
ULCC	Ultra Large Crude Carrier
VLCC	Very Large Crude Carrier

Exercise 5

- Good morning, this is Jan Nowak from the <Maritime Agency>. Could I speak to the <Chief Officer>, please.
- Speaking. What can I do for you?
- Your <request> for the inspection of <refrigerated holds> has been granted. The <surveyor> will come <on board> your vessel tomorrow at 15.00 hours. Will you please ask the <Second Engineer> to be present during the <inspection>.
- Yes, of course. He will be on board <expecting> the surveyor. Thank you for <arranging> the inspection so quickly as we have to load <refrigerated meat> the day after tomorrow and we need the required <certificates>. Thanks again and goodbye.
- Goodbye.

Exercise 6

- Harbour Master's Office, this is motor vessel "Nina" on <VHF channel> 12. We have completed <loading operations> this afternoon and we are due to leave this evening. I will need <two tugs> and <a pilot>. What time will the pilot be <available>? Over.
- Motor vessel "Nina", this is <Harbour Master> speaking. The pilot will <board> your vessel at 18.00 hours and the tugs will come at the same time to take you <out of the harbour> to the <anchorage>. Over.
- Harbour Master, this is <Captain> speaking. I shall <wait> for the pilot and the tugs at 18.00 hours. Thank you and see you in a <fortnight>. Over.
- Well, Captain. Have a good <voyage> and see you in a fortnight. Over and out.

Exercise 7

Tarifa VTS station, Tarifa VTS station, <Tarifa VTS station>, this is motor vessel Marina, motor vessel Marina, motor vessel Marina: <call sign> Alpha, Bravo, Charlie, Delta on VHF channel eight two. Over

Motor vessel Marina, motor vessel Marina, motor vessel Marina, this is Tarifa VTS station, Tarifa VTS station Tarifa VTS station. Question: What is your <position>? Over

Tarifa VTS station, Tarifa VTS station, Tarifa VTS station, this is motor vessel Marina call sign Alpha, Bravo, Charlie, Delta. Answer: My position is <10 miles west> of Tarifa VTS station.

Motor vessel Marina, This is Tarifa VTS station. <Understood> your position 10 miles west of Tarifa VTS. Question: What are your <course and speed>? Over.

Tarifa VTS station, this is motor vessel Marina, Answer: <my course> is 123 degrees, <speed> 10 knots. Over.

Motor vessel Marina, this is Tarifa VTS station. Question: What <type> of vessel are you and what is your <flag state>? Over.

Tarifa VTS station. This is motor vessel Marina. Answer: I am a Portuguese <reefer vessel>. Over

Motor vessel Marina, this is Tarifa VTS station. Question. What was your last <port of call>? Over

Tarifa VTS station, this is motor vessel Marina. Answer: My last <port of call> was Santa Cruz de Tenerife. Over.

Motor vessel Marina, This is Tarifa VTS station. Question: What is <your cargo>? Over.

Tarifa VTS station, this is motor vessel Marina. Answer: I have 3000 tons of <oranges> on board. Over.

Motor vessel Marina, this is Tarifa VTS station. Question: What is your <port of destination>? Over.

Tarifa VTS station, this is motor vessel Marina. Answer: My <port of destination> is Antwerp. Over

Motor vessel Marina. This is Tarifa VTS station. Have a <good voyage>. Over and out.

Exercise 8

1. Grain capacity of the vessel is <measured in cubic metres or feet>.
2. Crude oil can be carried in <tankers>.
3. Perishable goods include <fish and meat>.
4. Cargo space on board ro-ro ships can be expressed in <the length of lanes>.
5. Containers can also be carried <on deck>.
6. Japanese cars are usually transported by <PCC>.
7. Bulk cargoes include <coal>.
8. Liquefied natural gas is transported by <LNG vessels>.
9. Chemicals can be carried in <chemical carriers>.
10. Liquefied petroleum gas is transported by <LPG vessels>.

Exercise 9

Cargo space, as the name implies, is the **<space>** available for cargo to be carried **<on board>** different types of ships. It can be expressed as the **<grain capacity>** or **<bale capacity>** of the ship and can be found in the **<holds>**, in the 'tween-decks, in the **<tanks>** and deep tanks, in **<refrigerated>** compartments and **<cooling>** chambers and in the case of certain cargoes, such as **<timber>** or cargoes carried in **<refrigerated>** containers, even on deck.

Bale capacity is the **<cubic capacity>** of any space **<available>** for cargo such as bales of **<wool>** measured from the ceiling of the **<hold>** to the underside of the deck beams (the **<depth>**), between the inside of the cargo battens (the **<breadth>**) and between the inside of the **<bulkheads>** or sparring where fitted (the **<length>**). It is measured in cubic **<metres>** or **<cubic feet>**.

Grain capacity is the **<cubic capacity>** of any space available for cargo such as **<grain>** which fills the hold entirely. It is the total capacity in the hold with an allowance for the **<volume>** occupied by **<frames>** and beams. It is measured in cubic metres or **<cubic feet>**. It is **<greater>** than the bale capacity of the **<vessel>**.

Exercise 10

Broken **<stowage>** is the space between **<packages>** not taken up by the cargo, the space which remains **<unfilled>** for a variety of reasons such as the **<shape>** of the hold, the **<type>** of cargo, special **<kinds>** of packing, irregularly-sized items of **<machinery>** etc.

It is expressed as **<percentage>** when large **<cases>** have to be stowed in the hold.

Cargo space on board **<container>** vessels is usually measured and **<expressed>** in TEU which stands for **<Twenty-Foot>** Equivalent Unit.

Cargo space on board **<Ro-Ro>** ships is expressed in the length of **<lanes>** and is measured in **<metres>**. It can also be expressed as bale capacity and measured in cubic metres or cubic feet.

Cargo space on board Con-Ro vessels is expressed both in TEU for the number of containers the vessel can carry and in the number of cars it can take on the **<car>** deck so the **<length>** of lanes is also given.

Cargo space can be filled with **<different>** types of cargo. It can **<usually>** be divided into space for **<general>** cargo and **<bulk>** cargo, both liquid and **<dry>**, which is loose.

Different **<liquids>** and chemicals can evaporate from their **<receptacles>** and tanks and in such cases the unfilled space is called **<ullage>**.

Exercise 11

General cargo can be <bdivided> into <bcontainerised>, non-containerised and <brefrigerated> cargo. General cargo may cause many <bstowage> problems because the goods may be <bpacked> in different cases, <bbags>, boxes, <bbundles>, crates and <bdrums> and some pieces of <bmachinery> or heavy lifts can be loaded without any <bpackaging> at all. In such cases the <bbroken> stowage is very high.

Containerised cargo prevails nowadays and causes <bleast> problems in sea transport because <bcontainers> are of <bstandard> shape and dimensions.

<bPerishable> cargo is loaded into refrigerated containers or refrigerated holds and carried in <breefer> ships, which are specially <built> for that purpose. Perishable cargo includes <bmeat>, fish, <bdaairy> produce and <bfruit>.

Cars can be carried on board the PCC, which stands for <bPure Car> Carriers and cargo space in those ships is <bexpressed> in the length of <blanes> and the number of <bcars> they can carry.

Exercise 12

<bBulk> cargo can be divided into <bliquid> cargo and <bdry> cargo. Liquid cargo such as <bcrude> oil and its <bproducts> can be carried in <btankers>. Dry bulk cargo is usually carried by <bbulk> cargo vessels. <bOBO> vessels carry both types of bulk cargo. <bOBO> stands for Oil/Bulk/Ore.

LPG and LNG vessels carry <bLiquefied/liquefied> petroleum gas and <bLiquefied/liquefied> natural <bgas> in specially constructed <bsteel> spheres under <bpressure> and at <blow> temperatures. Chemicals such as <bmolten> sulphur are carried in special <bchemical> carriers.

Exercise 13

1. DECK
2. QUAY
3. DERRICK
4. DISCHARGE
5. BOOM
6. VESSEL
7. PIER

8. CARGO
9. CONTAINER
10. REEFER

Solution in the centre: CARGO SPACE

Exercise 14

Down:

1. CRATES
2. ANCHOR
3. ROLL-ON
4. GRAIN
5. ORANGE
6. CASES
7. AMOUNT
8. PETROL
9. ABOARD
10. CABIN
11. INWARD
12. TIMBER
13. YACHT

Solution at the top: CARGO CAPACITY

Exercise 15

What is the <deadweight> capacity of the vessel?

The <deadweight> capacity of the vessel is <5000> tonnes.

The hold capacity is <6000> cubic metres.

The container capacity is <1500> TEU.

The vessel will load <300> 20ft containers and <100> 40ft.

<10000> cubic metres of space are required.

The vessel can still load <500> tonnes.

How much <deck cargo> can the vessel load?

The vessel can load <100> 20ft containers

The vessel can load <twenty> trailers on the car deck.